

School of Computer Science and Engineering

MCA , 3rd Sem

**Internship Project Report
Data Analytics Intern**

**Organization Name-
Codec Technologies Pvt. Ltd.**

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Introduction

- **Overview:** Summarizes the industrial Artificial Intelligence internship conducted at Codec Technologies Pvt. Ltd.
- **Core Objective:** Build foundational understanding of AI, machine learning, deep learning, NLP, and Reinforcement Learning.
- **Practical Learning:** Python-based exercises, data preprocessing, model training, and evaluation.
- **Training Projects:** Graduate Admission Prediction using ANN and Handwritten Digit Classification using ANN.

Problem Statement

- **AI Learning Challenge:** Understanding how data is prepared and used to train AI and machine learning models.
- **Model Development:** Learning the basic workflow of selecting data, preparing inputs, training models, and evaluating results.
- **Neural Network Application:** Gaining practical exposure to ANN-based prediction and classification tasks.
- **Training Need:** Develop foundational skills through guided exercises in AI, deep learning, NLP, and Reinforcement Learning.

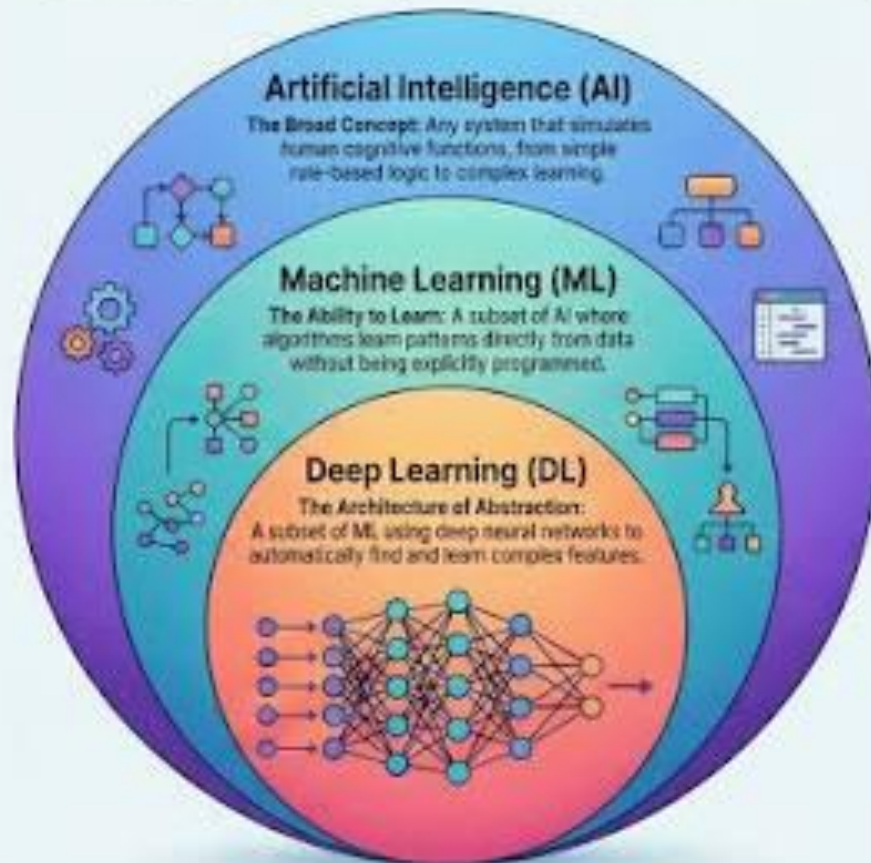
Scope of the Project

- **In-Scope Elements:**
 - Ingestion and validation of multi-attribute enterprise transactional records.
 - Automated Python routines for missing value imputation, duplicate purging, and schema casting.
 - Power Query ETL, Star Schema data modeling, and custom DAX metric calculations.
 - Multi-page interactive dashboards featuring dynamic slicers, drill-downs, and geospatial maps.
- **Out-of-Scope Elements:**
 - Real-time live database streaming connections (analysis performed on static enterprise exports).
 - Predictive machine learning algorithm deployment (focus remains strictly on Descriptive and Diagnostic analytics).

System Architecture

The Hierarchy of Modern Intelligence: AI, ML & DL Explained

Artificial Intelligence (AI) is the broad field of simulating human intelligence. Within AI sits Machine Learning (ML), which focuses on algorithms that learn from data. Deep Learning (DL) is a further specialized subset of ML that uses multi-layered neural networks to automatically learn complex patterns from raw data.



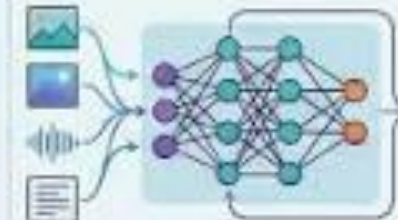
The Deciding Factor: How They Process Data

Traditional ML: Manual Feature Engineering



A human expert must manually select and create relevant data features for the algorithm to process.

Deep Learning: Automatic Feature Extraction



The neural network learns the most important features directly from raw data during training.

Practical Trade-Offs

Criterion	Traditional Machine Learning	Deep Learning
 Data Needs	 Effective with small to medium datasets.	 Requires very large datasets to perform well.
 Hardware	 Can run on standard CPUs.	 Needs high-performance GPUs or TPUs.
 Interpretability	 High ("White Box" - decisions are traceable).	 Low ("Black Box" - reasoning is opaque).

Technologies and Tools Used

- **Programming Language:** Python – used for AI training exercises.
Data Preparation: Preprocessing and visualization techniques.
- **Machine Learning:** Supervised learning and model evaluation concepts.
- **Deep Learning:** Artificial Neural Networks (ANN) and introductory CNN concepts.
- **NLP Tools:** NLTK and spaCy for basic text-processing tasks.
- **Reinforcement Learning:** Q-learning fundamentals.
- **Evaluation Metrics:** Accuracy, precision, recall, and F1-score.

Methodology

Phase 1 — AI & Data Foundations:

- Introduction to AI concepts and Python for AI development.
- Data preprocessing and visualization.

Phase 2 — Machine Learning:

- Supervised learning concepts.
- Model evaluation and optimization.

Phase 3 — ANN and Deep Learning Practical Work:

- Graduate Admission Prediction using ANN.
- Handwritten Digit Classification using ANN.
- Introduction to CNNs and basic image-classification model training.

Phase 4 — NLP and Reinforcement Learning:

- Text preprocessing and sentiment analysis using NLTK and spaCy.
- Implementation of a simple Q-learning model.
- Evaluation using accuracy, precision, recall, and F1-score.

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Thank You